

Cloud Computing for Real-Time Stock Market Prediction Using Machine Learning: A Cost to Benefit Review

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Abstract—Predicting stock market trends is essential for investors aiming to optimize returns. This paper examines the feasibility of using machine learning models and cloud computing to predict stock price movements based on news sentiment analysis. Meta, Google, Amazon, Apple, and Nvidia data was collected and processed using sentiment analysis and machine learning algorithms. Despite identifying some basic correlations between sentiment and stock movement, the models yielded inconclusive results, primarily due to limited training data. Furthermore, the study evaluates the cost-effectiveness of deploying such systems on cloud platforms and finds that the high infrastructure costs often outweigh potential returns. The findings underscore the challenges of applying real-time machine learning to financial forecasting, emphasizing the significance of balancing technical and economic considerations.

Index Terms—Stock market prediction, Cloud computing, Sentiment analysis, Cost-effectiveness, Real-time prediction

I. INTRODUCTION

Predicting stock market trends is a key goal for active investors who aim to optimize decisions based on metrics such as overall return, short-term gains, and public perception. However, stock values are impacted by a complex interplay of factors, including global events, market trends, and investor sentiment. To handle this complexity, machine learning has become an invaluable tool for analyzing the diverse and dynamic data points that influence stock performance.

A significant challenge lies in optimally extracting features from the vast amounts of data generated by the stock market. This enormous volume of information surpasses the capacity of a single machine, necessitating the adoption of scalable solutions like cloud computing. However, leveraging cloud infrastructure introduces additional costs that can impact the overall benefit. This paper explores the cost-to-benefit ratio of using cloud computing for real-time stock market prediction, analyzing the trade-offs between computational efficiency, scalability, and financial feasibility.

A. Paper overview

This paper is organized as follows: Section II provides a detailed review of related work, focusing on using machine learning for stock market prediction and the role of cloud computing in managing large-scale data processing. Section III outlines the methodology, including data collection, feature

extraction, and the deployment of machine learning models on cloud platforms. Section IV presents the results and analysis of the experiment. Then, Section V evaluates the cost-to-benefit trade-offs of different investment approaches. Finally, Section VI discusses the findings, their implications, and potential future work in this domain.

II. REVIEW OF LITERATURE

A. Predicting the Stock Market: Mission Impossible?

The Efficient Market Hypothesis (EMH) is the main principle of modern financial theory. It states that predicting the price of a stock using trend, fundamental, or technical analysis is impossible. However, the whole purpose of the hedge fund industry is to try to beat the market. Historically speaking, this sector of finance has not performed consistently well. Warren Buffett is famous for having bet with the Protégé Partners LLC hedge fund that an S&P 500 index fund would outperform a hand-picked portfolio of hedge funds over 10 years. Buffett won that bet [1].

While this part of history is not due to the scientific process, it still illustrates that trained human beings cannot efficiently predict the stock market. The question is now what is more valuable: investing into an ETF, using a financial advisor, or using machine learning with cloud computing. It is important to note that none of the papers used in this research mentions the cost of implementing their research for an individual.

B. Machine Learning for Stock Price Prediction

Researchers have acknowledged the challenge of accurately forecasting financial time series. This is due to economic conditions, political events, and trader expectations. The inherently nonlinear and irregular financial data also adds an extra layer of challenge [2]. However, some advancements have been made in the past decade to account for those challenges.

Machine learning is now being used to analyze and predict stock price movements in the long and short term. It can be helpful for long-term investing because analysts must evaluate many financial indicators to calculate a company's intrinsic value. Patil et al. [3] proposed a graph theory approach for forecasting stock prices, leveraging the spatiotemporal rela-

tionship between different stocks. They could model the stock market as a network by building two types of graphs: one based on the correlation of historical stock prices and the other based on the co-mention of stocks in financial news. They then used two hybrid models: deep learning convolutional neural networks (CNNs) and traditional machine learning techniques. The researchers used time series data for 30 stocks from the top 30 Fortune 500 companies and a news dataset containing one million financial news articles. They found that graph-based models outperform traditional statistical models and single linear models.

The paper “Equity Forecast: Predicting Long Term Stock Price Movement Using Machine Learning” [4] by Nikola Milosevic presents a methodology for predicting long-term stock price movements, specifically identifying whether a stock will grow by at least 10% within one year. The dataset was sourced from the Bloomberg terminal and includes 1739 stocks from indices like S&P 1000, FTSE 100, and S&P Europe 350. It contains financial indicators such as earnings per share (EPS), price-to-earnings ratio, and dividend yield. The machine learning algorithms used features such as book value, market capitalization, dividend yield, and price-to-book ratio. After feature selection, the Random Forest model achieved an F1 Score of 76.5%, outperforming other algorithms tested.

In the short-term stock value prediction, Rechenhth [5] offers interesting challenges to the EMH by examining whether conditional probabilities can be used to identify trends in high-frequency equity pricing. The paper explores the use of machine learning algorithms for predicting stock price direction in the near term (seconds to minutes). It reviews various machine learning algorithms, performance metrics, and methods to evaluate unseen data. The paper discusses the challenges of concept drift, imbalanced data streams, erroneous trade data, and dimensionality reduction in stock market prediction. The author then presents a new wrapper-based ensemble method incorporating technical analysis indicators, feature selection, and multiple data subsets to improve prediction accuracy. He suggests that using misclassification costs to optimize decisions, rather than relying solely on AUC, could be a promising area for future research.

Sun et al. [2] propose a novel AdaBoost-LSTM hybrid ensemble model for short-term financial time series forecasting, addressing challenges such as nonlinearity and irregularity in daily stock and exchange rate data. The method uses the AdaBoost algorithm to iteratively integrate multiple LSTM predictors, enhancing both level and directional forecasting accuracy. The dataset includes daily historical data for the S&P 500 Index, Shanghai Composite Index, EUR/USD, and USD/CNY exchange rates. Empirical results demonstrate that the AdaBoost-LSTM ensemble significantly outperforms traditional models and other approaches. They achieved a directional accuracy rates of 76.85% for EUR/USD out-of-sample predictions. This approach underlines the potential of hybrid ensemble learning in for short-term financial forecasting, particularly in high-frequency trading and decision-making scenarios.

Tipirisetty [6] explores a hybrid approach to predicting stock prices by combining numerical and textual analyses.

The author employs Long-Short-Term Memory (LSTM) networks to model stock trends as time-series data for numerical analysis and incorporates sentiment analysis of news headlines to capture textual influences. The sentiment scores of news headlines were derived using tools such as VADER and passed on to Support Vector Machines (SVM) and LSTMs to predict stock movement. Company-specific news headlines improved prediction accuracy significantly, with the best-performing LSTM model achieving over 80% accuracy in some cases.

III. METHODS

This paper presents a small-scale experiment that was conducted to evaluate the possibility of predicting the movement of various stocks using news sentiment analysis.

A. Data Collection and Pre-Processing

Effective stock market prediction relies on accurate and comprehensive data that captures market performance and external influencing factors. For this study, data, including current stock prices, trading volumes, and relevant news headlines, was gathered from Polygon.io and News API. We collected the values and headlines of Meta, Google, Amazon, Apple, and Nvidia from September 21st to October 22nd, 2024.

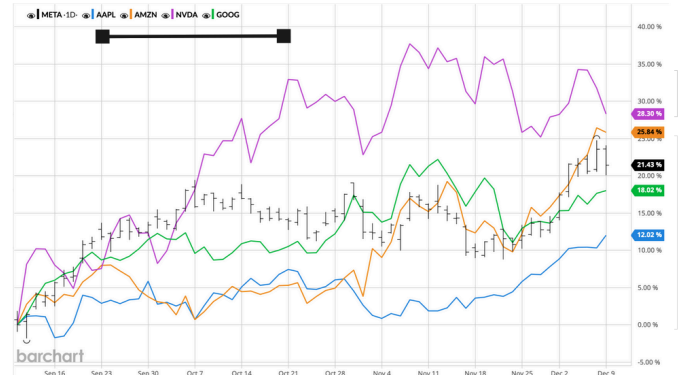


Fig. 1: The stock performance of each of the 5 companies and the timeframe corresponding to the study mentioned above.

The collected raw data was then stored in an InfluxDB Cloud database with the appropriate timestamp. Sentiment analysis was then performed on the news headlines using the Anthropic API. The prompt was to evaluate the sentiment of the headline on a scale from -10 to 10. The API was also tasked with rating irrelevant headlines with a score of 101. Then, the stored headlines were filtered by rating. The movement of the stock was evaluated by percentage, where each value was calculated as

$$\frac{\text{Closing Price} - \text{Opening Price}}{\text{Opening Price}} * 100\% \quad (1)$$

This calculation was performed on the hourly and daily value of the stock.

B. Machine Learning Modeling

Two machine learning models were used to predict stock performance:

- Regression Model: Linear regression was applied to predict numerical changes in stock prices.
- Classification Model: Logistic regression classified stocks into categories based on percentage changes.

The data was split into training and testing sets, with the training sets (80% of the data) used for fitting the models and the testing sets (20% of the data) used for evaluation.

C. Evaluation Metrics

The models were evaluated using the following metrics:

- Regression: Mean Squared Error (MSE) and R-squared were used to assess the accuracy of numerical predictions.
- Classification: Accuracy was used to evaluate categorical predictions.

IV. RESULTS

The linear regression results were inconclusive for both the hourly and daily data. The predicted price difference does not vary enough with the current model. The problem might be the limited amount of training data available for the experiment. The only accurate part of the results is that positive and negative price differences are in the correct overall category.

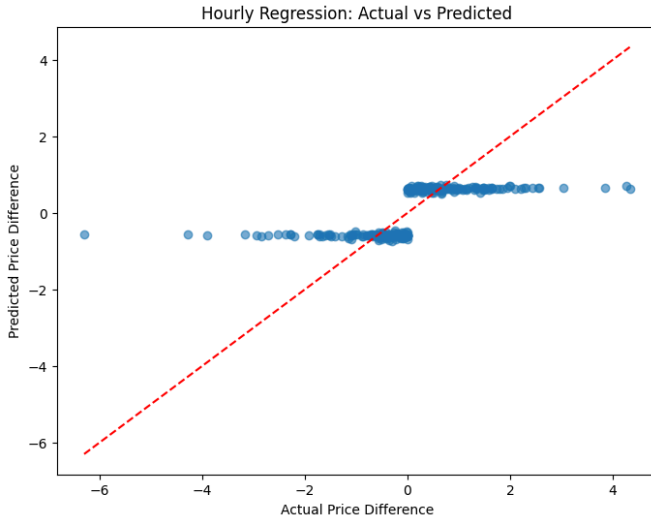


Fig. 2: The linear regression results for the hourly regression

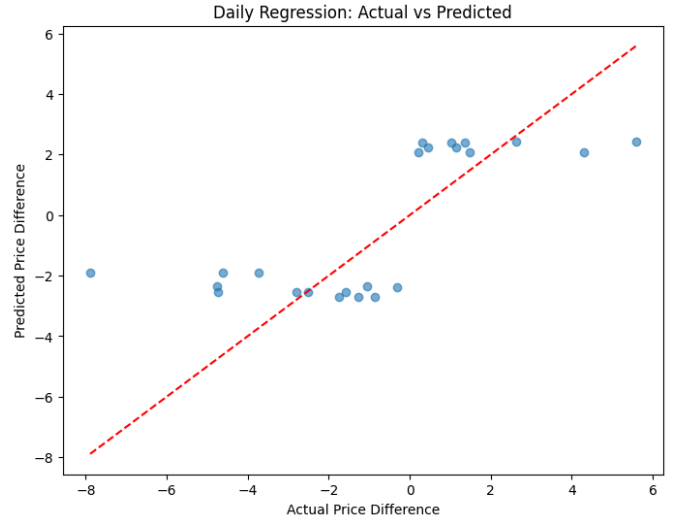


Fig. 3: The linear regression for the daily regression

The classification results were also inconclusive, with the hourly classification having an accuracy of 62% and the daily classification having an accuracy of 57%. Once again, the limited training data might be the main reason the classification fails. We can see that the overall idea of price being up or down can correlate with the prediction. The main question is the appropriate amount of data to push this prediction further.

TABLE I: RESULTS FOR HOURLY CLASSIFICATION

Class	Precision	Re-call	F1-Score	Support
0	0.60	0.62	0.61	186
1	0.63	0.61	0.62	196
Accuracy	-	-	0.62	382
macro avg	0.62	0.62	0.62	382
weighted avg	0.62	0.62	0.62	382

TABLE II: RESULTS FOR DAILY CLASSIFICATION

Class	Precision	Re-call	F1-Score	Support
0	0.50	0.20	0.29	10
1	0.58	0.85	0.69	13
Accuracy	-	-	0.57	23
macro avg	0.54	0.52	0.49	23
weighted avg	0.54	0.57	0.51	23

V. DISCUSSION

A. The price of predicting the stock market

Predicting the stock market can be tedious and involve a significant number of potential pitfalls. The main problem is that the funds to allocate to this project, both the original investment and the cost of running the system, can be very high. For example, the TradeStation API (to make online trades) requires an original investment of \$10,000 to get an API key. Then, running a virtual machine to collect news articles, stock data, process the data, run the machine learning model, and make the trades could cost about \$450/month (Google Cloud N2 with 16 vCPU and 64 GB memory). A real-time news API and stock API would increase the price by \$400/month (\$200 for the News API and \$200 for the Polygon API). The LLM cost for the news sentiment analysis is more complicated to predict since the number of tokens and the price per token can vary. For simplicity, let's assume that the individual uses ChatGPT Pro at \$200/month. Finally, we also need to account for the cloud database. The cost could be fixed at \$50/month. This leads to an original investment of \$10,000 and a monthly fee of at least \$1100. This means a potential loss of \$49,600 if the algorithm proves useless.

In comparison, investing \$10,000 in an S&P 500 ETF (VOO is used for the calculation) in January 2022 and adding \$1100/month to the investment would give an ending investment balance of \$60,874.18 after 3 years with the average annual rate of return of 11.47%. This means an earning of \$548.35, not counting dividends.

This solution is also only available to individual traders. AI contributing to trading might not benefit Wall Street's model. It is unclear if multiple AI making trades could potential leads to a crash.

VI. CONCLUSION

This study highlights the challenges and limitations of using machine learning models to predict stock market trends in real-time, particularly when constrained by limited data and high computational costs. While the results indicate a basic correlation between sentiment and stock price movement, they

fall short of producing reliable predictions, probably due to the small dataset used in this experiment.

The financial feasibility of implementing a cloud-based machine learning system for individual investors also presents significant hurdles. The cost of cloud infrastructure, APIs, and sentiment analysis tools can quickly exceed the returns generated by the predictions. Furthermore, when compared to a traditional investment strategy, such as a steady investment in an S&P 500 ETF, the machine learning approach appears less financially viable for most individual traders.

Future work should focus on acquiring larger, more diverse datasets and refining feature extraction techniques to improve model performance. Additionally, exploring alternative machine learning architectures and optimizing resource allocation on cloud platforms could enhance both accuracy and cost-effectiveness. While promising, this approach currently requires significant advancements to justify its adoption over more traditional investment strategies.

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